Ideation Phase
Empathize & Discover

Date	31 January 2026
Project Name	**Visualization Tool for Electric Vehicle Charge and Range Analysis**
Maximum Marks	4 Marks

Empathy Map Canvas:

An empathy map is a simple, easy-to-digest visual that captures knowledge about a user's behaviors and attitudes. It is a useful tool that helps teams better understand their users. Creating an effective solution requires understanding the true problem and the person who is experiencing it. The exercise of creating the map helps participants consider things from the user's perspective along with their goals and challenges.

Target User: Electric Vehicle (EV) Driver

What does he/she THINK AND FEEL?

- **"Will I have enough battery to make it to my destination and back?"**
- **"I hope the public charging station isn't broken or already occupied."**
- **Preoccupied with "range anxiety" during long trips.**
- **Feels environmentally responsible but frustrated by the lack of cohesive planning tools.**

What does he/she HEAR?

- **News about new EV infrastructure bills and charging networks being built.**
- **Friends saying, "I'd buy an EV, but charging on road trips seems like a hassle."**
- **Other EV drivers complaining about fragmented charging apps and payment methods.**

What does he/she SEE?

- **The car's dashboard battery percentage slowly dropping.**
- **Multiple, disconnected apps on their phone just to find different brands of charging stations.**
- **Gas stations everywhere, but public EV chargers are often hidden or poorly marked.**
- **A confusing market offering different plug types (CCS, CHAdeMO, Tesla).**

What does he/she SAY AND DO?

- **Spends 20+ minutes meticulously planning routes for road trips to include charging stops.**
- **Complains on forums about unreliable public charging infrastructure.**
- **Always unplugs and plugs in the car at home as part of a daily routine.**
- **Checks the weather and elevation, knowing it might drain the battery faster.**

Pain

- **Range anxiety and the fear of getting stranded.**
- **Wasting time waiting in line at occupied charging stations.**
- **The cognitive load of using multiple apps to calculate routes, check station status, and pay.**
- **Obstacles like broken chargers or "ICE-ing" (gas cars parked in EV spots).**

Gain

- **Wants a single, unified dashboard (like your Tableau tool) to see an accurate driving radius.**
- **Needs real-time data on charging station availability and compatibility.**
- **Measures success by completing trips without stress, saving time, and saving money on fuel.**
- **Desires peace of mind and confidence to travel anywhere.**